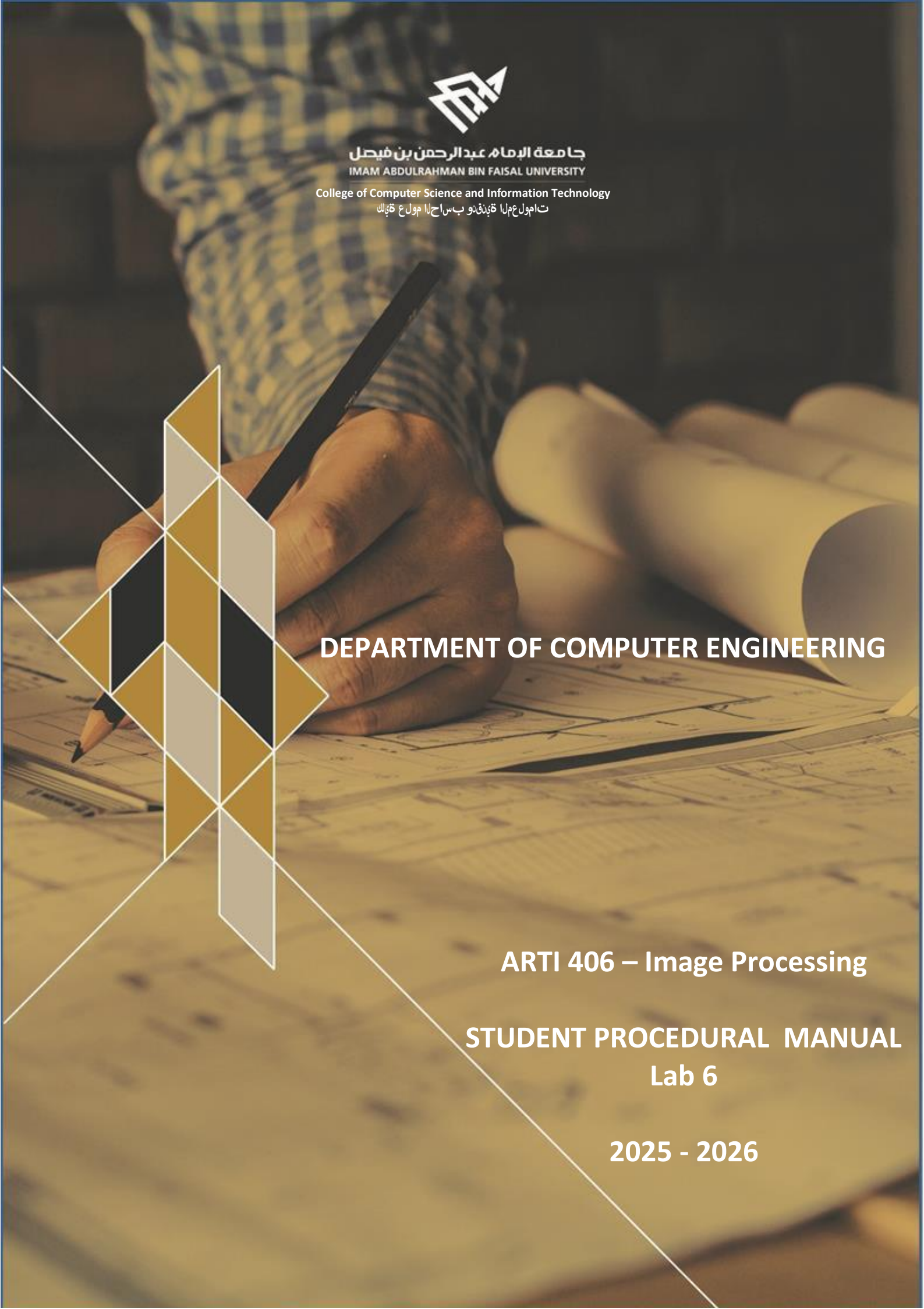


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DEPARTMENT OF COMPUTER ENGINEERING

ARTI 406 – Image Processing

STUDENT PROCEDURAL MANUAL
Lab 6

2025 - 2026

IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

Lab 6 – Answers: Frequency Domain

Task: Sobel Filters in the Frequency Domain

Steps:

1. Take the FFT of the Cameraman image.

2. Take the FFT of the Sobel X and Sobel Y kernels (padded to the image size).

3. Multiply the image FFT by each filter FFT.

4. Take the inverse FFT to get the result back as an image.

5. Combine Sobel X and Sobel Y to get the edge magnitude.

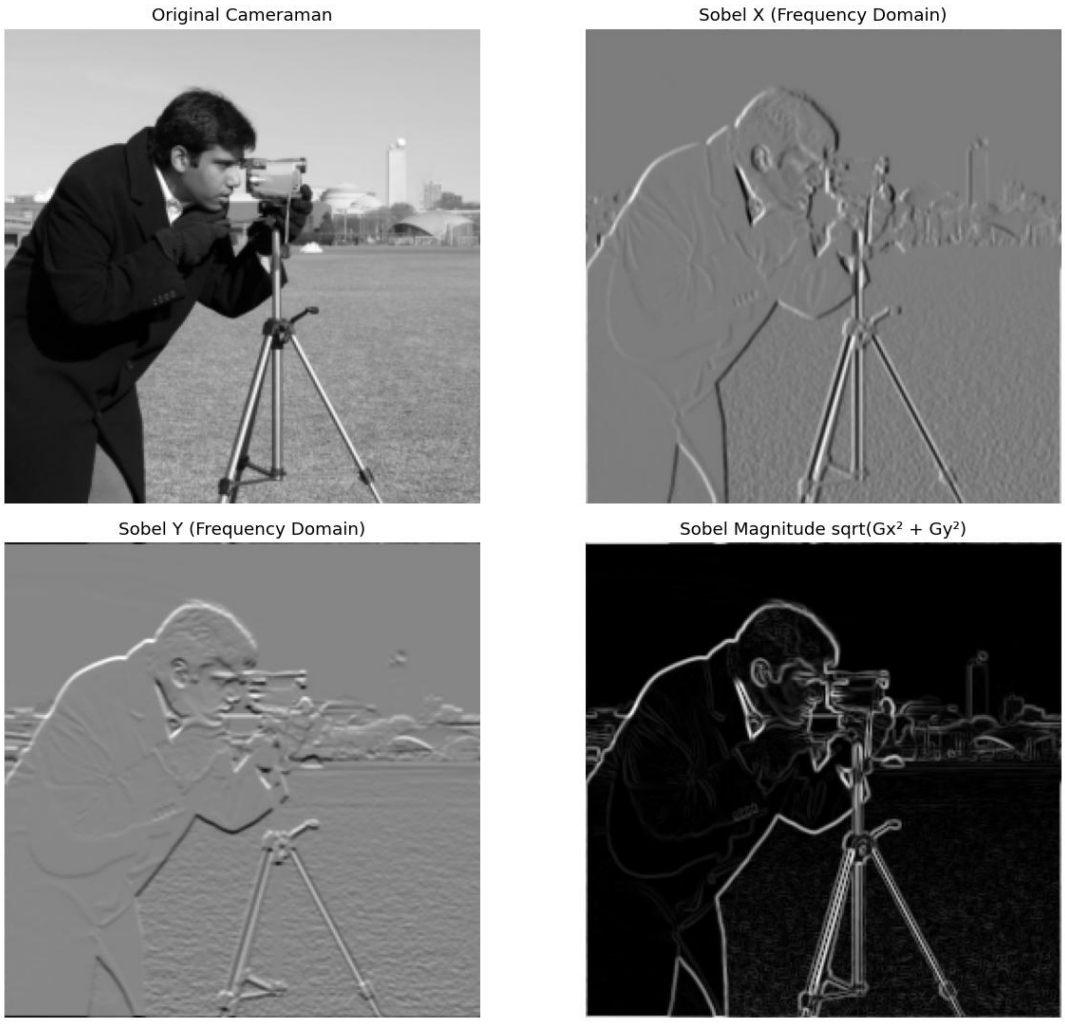


Figure 1: Sobel filters applied in the frequency domain

Explanation

Sobel X: finds the vertical edges, like the tripod legs and the outline of the coat.

Sobel Y: finds the horizontal edges, like the horizon, the top of the camera and the shoulders.

Magnitude: combines both directions and shows all the edges in the image.

The result is the same as applying Sobel directly on the image (the difference is almost 0), because multiplying in the frequency domain is the same as convolution in the spatial domain.